

Useful Identities

We will prove a few identities used by Jackson, whose proofs are not provided.

$$\mathbf{1} \quad \int_V \nabla \psi \, d^3x = \int_S \psi \mathbf{n} \, da$$

To prove this identity, notice that

$$\mathbf{c} \cdot \nabla \psi = \nabla \cdot (\psi \mathbf{c}) - \psi \nabla \cdot \mathbf{c} = \nabla \cdot (\psi \mathbf{c}),$$

for an arbitrary constant vector $\mathbf{c}$. Then, applying Gauss theorem, we have

$$\mathbf{c} \cdot \int_V \nabla \psi \, d^3x = \int_V \nabla \cdot (\psi \mathbf{c}) \, d^3x = \int_S (\psi \mathbf{c}) \cdot \mathbf{n} \, da = \mathbf{c} \cdot \int_S \psi \mathbf{n} \, da.$$

Since $\mathbf{c}$ is arbitrary, we must have

$$\int_V \nabla \psi \, d^3x = \int_S \psi \mathbf{n} \, da.$$

This identity is used to transform Eq. (4.14) to Eq. (4.15).

$$\mathbf{2} \quad \int_V \nabla \times \mathbf{A} \, d^3x = \int_S \mathbf{n} \times \mathbf{A} \, da$$

Again, for arbitrary constant vector $\mathbf{c}$,

$$\mathbf{c} \cdot (\nabla \times \mathbf{A}) = \nabla \cdot (\mathbf{A} \times \mathbf{c}) + \mathbf{A} \cdot (\nabla \times \mathbf{c}) = \nabla \cdot (\mathbf{A} \times \mathbf{c}).$$

Applying Gauss theorem,

$$\mathbf{c} \cdot \int_V \nabla \times \mathbf{A} \, d^3x = \int_V \nabla \cdot (\mathbf{A} \times \mathbf{c}) \, d^3x = \int_S \mathbf{n} \cdot (\mathbf{A} \times \mathbf{c}) \, da = \mathbf{c} \cdot \int_S (\mathbf{n} \times \mathbf{A}) \, da.$$

Since $\mathbf{c}$ is arbitrary, we must have

$$\int_V \nabla \times \mathbf{A} \, d^3x = \int_S \mathbf{n} \times \mathbf{A} \, da.$$

This identity is applied to Eq. (5.60) to convert the integration over the entire space to a surface integral.

$$3 \quad \int_S \mathbf{n} \times \nabla \psi \, da = \oint_C \psi \, d\mathbf{l}$$

Again, for arbitrary constant vector $\mathbf{c}$,

$$\mathbf{c} \cdot (\mathbf{n} \times \nabla \psi) = \mathbf{n} \cdot (\nabla \psi \times \mathbf{c}) = \mathbf{n} \cdot [\nabla \times (\psi \mathbf{c}) - \psi (\nabla \times \mathbf{c})] = \mathbf{n} \cdot [\nabla \times (\psi \mathbf{c})].$$

Applying Stokes theorem,

$$\mathbf{c} \cdot \int_S (\mathbf{n} \times \nabla \psi) \, da = \int_S \mathbf{n} \cdot [\nabla \times (\psi \mathbf{c})] \, da = \oint_C (\psi \mathbf{c}) \cdot d\mathbf{l} = \mathbf{c} \cdot \oint_C \psi \, d\mathbf{l}.$$

Since $\mathbf{c}$ is arbitrary, we must have

$$\int_S \mathbf{n} \times \nabla \psi \, da = \oint_C \psi \, d\mathbf{l}.$$

$$4 \quad \oint_C \mathbf{A} \times d\mathbf{l} = \int_S (\nabla \cdot \mathbf{A}) \mathbf{n} \, da - \int_S (\nabla \mathbf{A}) \cdot \mathbf{n} \, da$$

For arbitrary constant vector $\mathbf{c}$,

$$\mathbf{c} \cdot \oint_C \mathbf{A} \times d\mathbf{l} = \oint_C (\mathbf{c} \times \mathbf{A}) \cdot d\mathbf{l} = \int_S [\nabla \times (\mathbf{c} \times \mathbf{A})] \cdot \mathbf{n} \, da.$$

Since

$$\nabla \times (\mathbf{c} \times \mathbf{A}) = \mathbf{c} (\nabla \cdot \mathbf{A}) - \mathbf{A} (\nabla \cdot \mathbf{c}) + (\mathbf{A} \cdot \nabla) \mathbf{c} - (\mathbf{c} \cdot \nabla) \mathbf{A} = \mathbf{c} (\nabla \cdot \mathbf{A}) - (\mathbf{c} \cdot \nabla) \mathbf{A},$$

we will have

$$\mathbf{c} \cdot \oint_C \mathbf{A} \times d\mathbf{l} = \int_S (\nabla \cdot \mathbf{A}) (\mathbf{c} \cdot \mathbf{n}) \, da - \int_S (\mathbf{c} \cdot (\nabla \mathbf{A}) \cdot \mathbf{n}) \, da = \mathbf{c} \cdot \left[\int_S (\nabla \cdot \mathbf{A}) \mathbf{n} \, da - \int_S (\nabla \mathbf{A}) \cdot \mathbf{n} \, da \right],$$

or

$$\oint_C \mathbf{A} \times d\mathbf{l} = \int_S (\nabla \cdot \mathbf{A}) \mathbf{n} \, da - \int_S (\nabla \mathbf{A}) \cdot \mathbf{n} \, da.$$

In particular, for $\mathbf{A} = \mathbf{x}$, we know

$$\nabla \cdot \mathbf{x} = 3, \quad (\nabla \mathbf{x}) \cdot \mathbf{n} = \sum_i \sum_j \hat{e}_j \frac{\partial x_i}{\partial x_j} n_i = \mathbf{n}.$$

Then,

$$\oint_C \mathbf{x} \times d\mathbf{l} = \int_S [3\mathbf{n} - \mathbf{n}] \, da = \int_S 2\mathbf{n} \, da = 2S\mathbf{n},$$

where S is the area of the surface. This established Eq. (5.57).

$$5 \quad \int_V \mathbf{A} \, d^3x = - \int_V \mathbf{x} (\nabla \cdot \mathbf{A}) \, d^3x$$

Consider the following obvious identity,

$$\nabla \cdot (x_i \mathbf{A}) = (\nabla x_i) \cdot \mathbf{A} + x_i (\nabla \cdot \mathbf{A}) = A_i + x_i (\nabla \cdot \mathbf{A}).$$

Then,

$$\sum_{i=1}^3 \hat{e}_i [\nabla \cdot (x_i \mathbf{A})] = \mathbf{A} + \mathbf{x} (\nabla \cdot \mathbf{A}).$$

Integrate over the entire space and apply the Gauss theorem, the left hand side becomes

$$\sum_{i=1}^3 \hat{e}_i \int_V \nabla \cdot (x_i \mathbf{A}) \, d^3x = \sum_{i=1}^3 \hat{e}_i \int_S \mathbf{n} \cdot (x_i \mathbf{A}) \, da.$$

If $\mathbf{A}$ is localized and drops to zero at infinity, or decreases fast enough, $|\mathbf{A}| = o(1/r^3)$ for $r \rightarrow \infty$, the surface integral will be zero, which leaves

$$\int_V [\mathbf{A} + \mathbf{x} (\nabla \cdot \mathbf{A})] \, d^3x = 0,$$

or,

$$\int_V \mathbf{A} \, d^3x = - \int_V \mathbf{x} (\nabla \cdot \mathbf{A}) \, d^3x$$

$$6 \quad \det(\delta_{ij} + a_i b_j) = 1 + \mathbf{a} \cdot \mathbf{b}$$

This identity is not mentioned in Jackson, but can be useful for Prob. 6.2.

Notice that,

$$\begin{pmatrix} F & C \\ B & D \end{pmatrix} = \begin{pmatrix} I_n & C \\ 0 & D \end{pmatrix} \begin{pmatrix} F - CD^{-1}B & C \\ d^{-1}B & I_m \end{pmatrix} = \begin{pmatrix} F & 0 \\ B & I_m \end{pmatrix} \begin{pmatrix} I_n & F^{-1}C \\ 0 & D - BF^{-1}C \end{pmatrix},$$

where F is an invertible $n \times n$ matrix, B an $m \times n$ matrix, C an $n \times m$ matrix, D an invertible $m \times m$ matrix, and I_k a $k \times k$ identity matrix. Then,

$$\det \begin{pmatrix} F & C \\ B & D \end{pmatrix} = \det \begin{pmatrix} I_n & C \\ 0 & D \end{pmatrix} \cdot \det \begin{pmatrix} F - CD^{-1}B & C \\ d^{-1}B & I_m \end{pmatrix} = \det D \det (F - CD^{-1}B),$$

and

$$\det \begin{pmatrix} F & C \\ B & D \end{pmatrix} = \det F \det (D - BF^{-1}C).$$

Now, let

$$F = I_n, \quad D = I_1 \equiv 1, \quad B = (b_1, \dots, b_n), \quad C = -(a_1, \dots, a_n)^\top,$$

then we have

$$\det \begin{pmatrix} I_n & C \\ B & I_1 \end{pmatrix} = \det I_n \det (1 - BC) = 1 + \sum_i a_i b_i,$$

and also

$$\det \begin{pmatrix} I_n & C \\ B & I_1 \end{pmatrix} = \det I_1 \det (I_n - CB) = \det (\delta_{ij} + a_i b_j).$$

Therefore, we have established

$$\det (\delta_{ij} + a_i b_j) = 1 + \mathbf{a} \cdot \mathbf{b}.$$